

Weekly Report

iOS Application Development

Date: April 21 – April 28, 2026
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Projects: Employ210 & SNCC

1. AWS S3 Region Migration (Employ210)

A critical infrastructure issue was identified and resolved this week. The AWS Amplify application was deployed in the **us-east-2** (Ohio) region, while the S3 storage bucket was located in **us-east-1** (N. Virginia). This mismatch was causing storage operations to fail and preventing proper integration with Lambda functions being developed by the team.

Resolution Steps:

- **Created New S3 Bucket:** A new S3 bucket named *employ210-storage-east2* was created in the us-east-2 region to match the Amplify deployment.
- **Updated Configuration Files:** Both *awsconfiguration.json* and *amplifyconfiguration.json* were updated to point to the new bucket with the correct region settings.
- **Configured Bucket Permissions:** Appropriate public access settings and IAM policies were configured to allow the iOS application to upload HTA reports and video recordings.
- **Team Coordination:** Coordinated with teammate Yassine to ensure all future AWS resources (Lambda functions, pipelines) will be created in us-east-2 for consistency.

2. AWS Configuration & Debugging (Employ210)

Significant debugging effort was invested in resolving persistent **Amplify.StorageError error 1** (Access Denied) issues that were occurring on physical iOS devices while working correctly in the Simulator.

Issues Identified & Fixed:

- **AppClientId Mismatch:** The two configuration files had different Cognito App Client IDs, causing authentication inconsistencies. Both files were synchronized to use the same credentials.
- **Enhanced Error Logging:** Detailed error logging was added to the S3 upload functions to capture StorageError descriptions, recovery suggestions, and underlying errors for faster debugging.
- **IAM Policy Updates:** Added S3 access policies to both authenticated and unauthenticated Cognito Identity Pool roles to ensure proper permissions for storage operations.

3. SNCC App - Prompt Engineering

Continued development on the SNCC (behavioral classification) application focused on improving the accuracy and reliability of AI-powered analysis.

Work Completed:

- **Prompt Refinement:** Edited the AI prompts to return exact confidence numbers for behavioral classification categories (A through E), improving the precision of the analysis output.
- **Case Review with Katie:** Met with Katie to review real-world video examples collected from the field. Analyzed cases where the current prompts were failing to correctly classify behaviors.
- **Ongoing Fixes:** Currently working on prompt adjustments to address the identified failure cases and improve overall classification accuracy.

4. Collaboration & Meetings

- Met with **Yassine** to discuss AWS infrastructure alignment and confirm all resources will be in us-east-2 region
- Met with **Katie** to review SNCC video examples and identify prompt failure patterns

Upcoming Tasks

Priority	Task	Project
High	Fix SNCC prompt failures identified with Katie	SNCC
High	Test S3 uploads on physical device with new bucket	Employ210
Medium	Implement new UI pages discussed with Katie	Employ210
Medium	Begin AI video analysis integration	Employ210